Granite Tait · Oct 28, 2026

Terra

A custom AI model that **moves AI work off the grid's rush hours.**

Scan now. It's live.

terra.granitetait.com

github.com/granman3/terra

THE PROBLEM

$1.1 trillion

what US power companies plan to spend on the grid in **just five years**, 2025–29, the most ever. Peak demand is outrunning the grid, and **data centers are the main new driver of that demand.**

DEMAND

+224 GW

new US peak demand in 10 years. 13 of 23 regions now face a risk of running short.

AI'S SHARE

4.4% → 12%

of all US electricity going to data centers, 2023 → 2028

WATER

211B gal

of water a year just to make data centers' power, 12× what they use on site

WASTED FUEL

+40%

more gas per kWh burned by the peaker plants that cover the worst hours

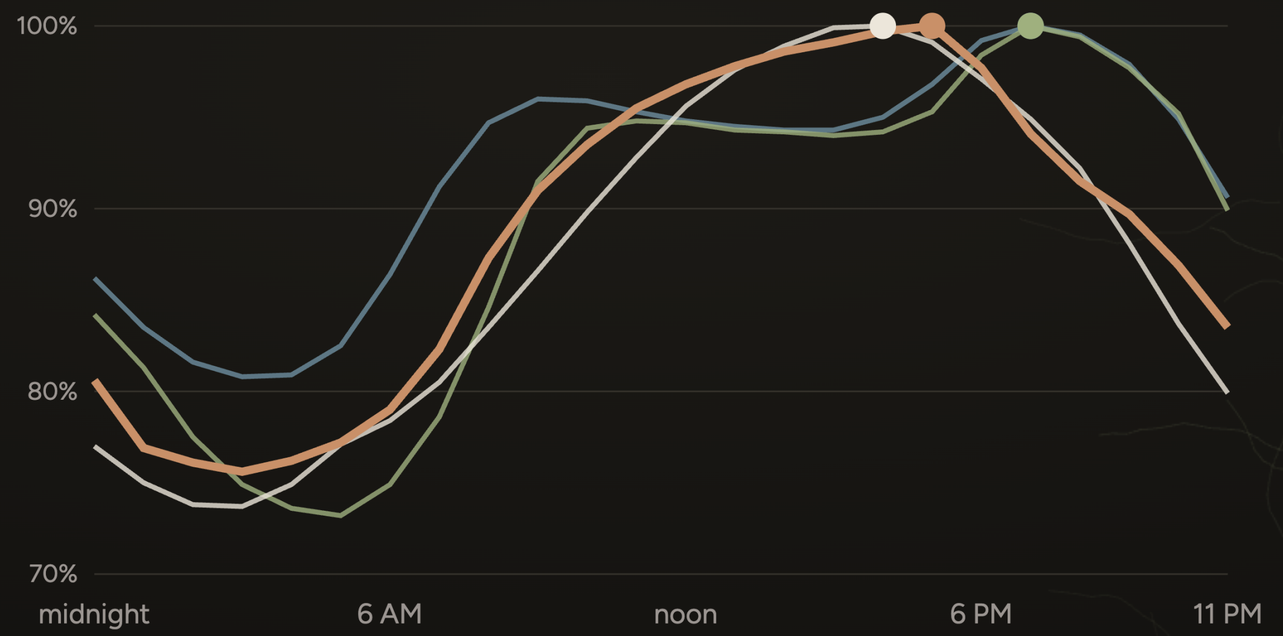
In Utah, the top 10% of the grid, **972 MW**, was needed for only **153 hours** in 2025: **$0.7–1.9 billion** of plants idle 98% of the year.

Why it's so hard: every load is treated as fixed, so the grid gets built for its worst hours. **Terra makes the biggest new load move.**

EEI 2025 · NERC 2025 LTRA · DOE/LBNL Dec 2024 (water, 2023) · EIA heat rates · EIA-930, PACE 2025 · GridLab 2025 peaker costs

THE SOLUTION, AND WHY I PICKED IT

Power plants can't move. AI work can.



Average demand through the day, 2026, each grid vs its own high (Utah time)

85%

of Utah's rush hours in 2026, California, the Pacific NW or Texas had room to spare

Build peaker plants	$728–1,969 / kW
Batteries (Torus Pulse)	for load that can't move
Move the load (Terra)	$0 to build

98 GW of new data centers fit on today's US grid if they flex 0.5% of the time. 42% of AI chip time can wait.

EIA-930, 2026, rush hour = demand ≥ 95% of that day's high · GridLab 2025 · Duke Nicholas Institute, Feb 2025 · Alibaba 2026 trace

WHAT I BUILT

A custom AI model, trained on my own RTX 4080.

33 million public data points → 1.04 million examples → 5 training runs, 3 designs → 561,000-parameter model → Torus Lasso

THE MATH: EVERY JOB, EVERY HOUR

$$(r^*,\ s^*) = \arg\min_{r,\ s+4\le D} \sum_{h=s}^{s+3} \left(\hat{p}_{r,h} + \frac{g_{r,h}}{4000}\right)$$

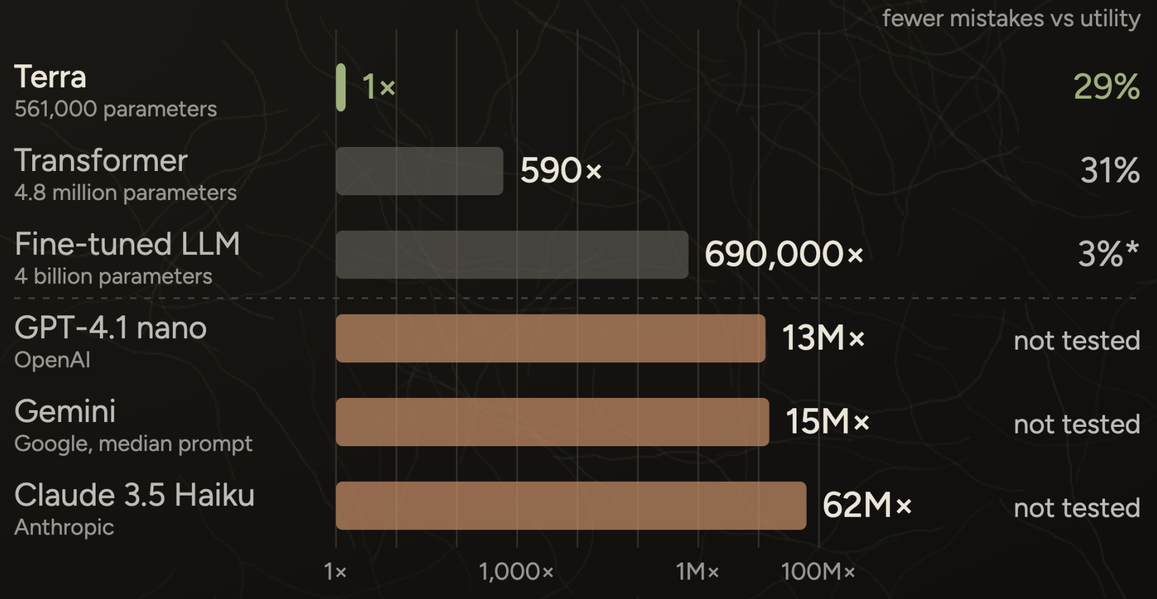
Pick the region **r** and start hour **s** where the job's 4 hours have the **least chance of rush hour** ($\hat{p}$, Terra's forecast). Carbon breaks ties.

29%
fewer mistakes than the utility's own forecast

1.14 → 0.04
rush hours a 4-hour AI job runs into

Why small: energy per decision

times Terra's, log scale (each line is 10×) · copper: small frontier models, published figures



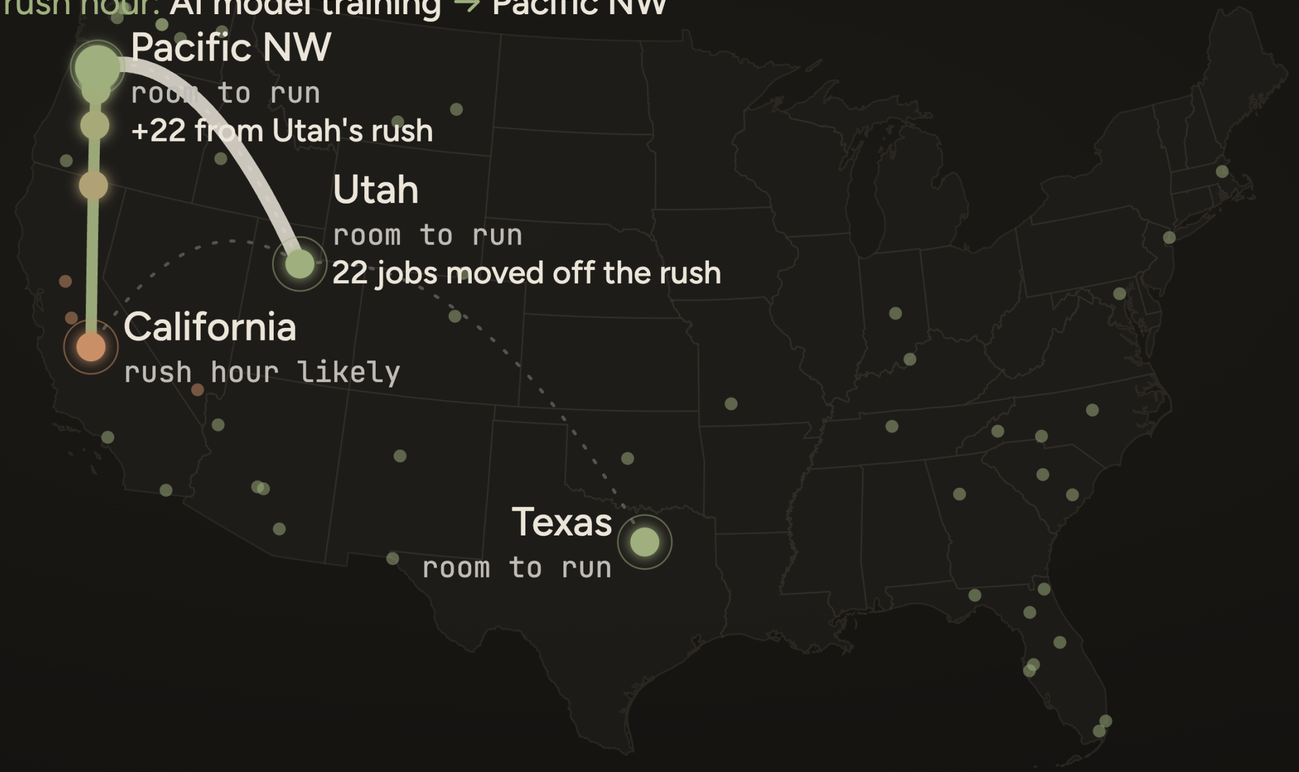
held-out 2026 vs EIA (Brier) · my models: same laptop · frontier: published Wh per query, arXiv 2505.09598 + Google 2025 · *4-region test

50 regions forecast · 4 expecting rush hour

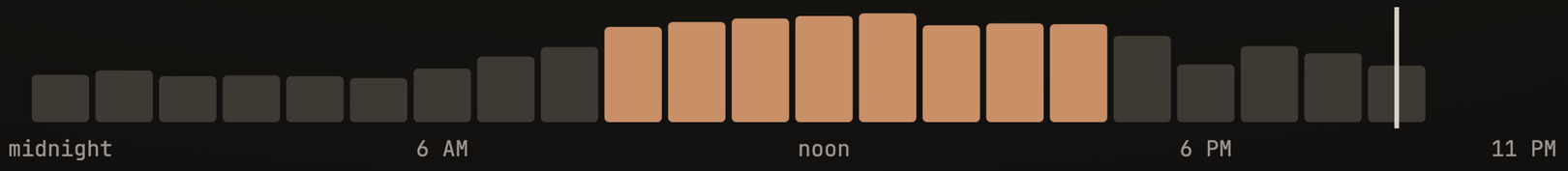
Utah has room

Power use at 81% of today's high · Utah takes work from grids in their rush hour

California rush hour: AI model training → Pacific NW



UTAH TODAY: POWER USE BY HOUR · COPPER = RUSH HOUR · EACH DOT ABOVE = ONE AI JOB MOVED OFF IT



Real EIA grid data, Oct 19–25, 2026, unseen in training · real Utah prices (CAISO) · the data center and its jobs are simulated

MOVED OFF UTAH'S RUSH HOUR

22 AI jobs sent to grids with room to spare

California	0 jobs
Pacific NW	22 jobs
Texas	0 jobs

counted only during Utah's rush hours · this week so far

WHAT UTAH SAVES

$10M a year

per **100 MW** of AI work moved off Utah's rush hours

THIS WEEK · ONE SIMULATED SITE

Power moved off the rush	495 kWh
Saved so far	$44.87

WHY IT MATTERS: UTAH, EVERY GRID, AND TORUS

TORUS SPIN

Milliseconds

Flywheel catches the fast swings.

TORUS PULSE

Hours

Battery covers the long rush hour.

TERRA

The load itself

Moves the demand: to a region with room, or a later hour. A third asset for Lasso, nothing new to build.

UTAH

$10M a year

for every 100 MW of AI work that can wait: 100,000 kW × $100 per kW a year, Rocky Mountain Power's own price

EVERY GRID

98 GW

of new AI load today's US grid can take with no new plants, once it flexes off the peaks

POWER SPIKES

813 → 16 kW

a 1 MW AI site's one-second jump with staggered starts. Less for Spin to catch.

HOW WE GET THERE

Built

Forecast the rush, a day ahead

50 regions. Utah: 80% of rush hours caught vs 69% for the utility.

Next

Plug into Lasso

Pilot with one Utah AI site on Rocky Mountain Power's demand-response program.

Then

Every GPU on the grid

Utah's regulators have a docket open on a flexible large-load tariff (PSC 24-035-43).

Wattsmart Business demand response, Rocky Mountain Power · Utah Code 54-26 (SB 132, 2025), PSC docket 24-035-43 · rush hours caught: held-out 2026 data